

Third Semester B.Tech Degree Examination Dec. 2021
MAT 203

PART A

1 Show that $p \wedge (\neg p \wedge q)$ is a contradiction. Use truth table

A:

P	q	$\neg p$	$\neg p \wedge q$	$p \wedge (\neg p \wedge q)$
0	0	1	0	0
0	1	1	1	0
1	0	0	0	0
1	1	0	0	0

2 Show the following implication without constructing a truth table.

$$(P \wedge Q) \Rightarrow P \rightarrow Q$$

A:

Steps

Reasons

1) $P \wedge Q$

Premise

2) Q

conjunctive simplification

3) $\neg p \vee Q$

disjunctive amplification

4) $\therefore P \rightarrow Q$

$$\neg p \vee Q \Leftrightarrow P \rightarrow Q$$

3. Find the number of permutations 1, 2, 3, 4, 5, 6, 7 that are not derangements?

A: Total number of permutations = $7! = 5040$

$$\begin{aligned}\text{No. of derangements} = D_7 &= 7! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} \right] \\ &= \frac{7!}{e} = 1854\end{aligned}$$

$$\begin{aligned}\therefore \text{The no. of permutations that are not derangements} \\ &= 5040 - 1854 = 3186.\end{aligned}$$

4. Prove (i) $nC_0 + nC_1 + nC_2 + \dots + nC_n = 2^n$

(ii) $nC_0 - nC_1 + nC_2 - \dots + (-1)^n nC_n = 0$, where n is a positive integer.

A: If x and y are variables and n is a positive integer, then

$$(x+y)^n = nC_0 x^0 y^n + nC_1 x y^{n-1} + nC_2 x^2 y^{n-2} + \dots + nC_n x^n y^0. \quad \text{--- ①}$$

(i) Let $x=1, y=1$, then eqn ① becomes

$$(1+1)^n = nC_0 1^0 1^n + nC_1 1^1 1^{n-1} + nC_2 1^2 1^{n-2} + \dots + nC_n 1^n 1^0$$

$$2^n = nC_0 + nC_1 + nC_2 + \dots + nC_n.$$

(ii) Let $x=-1, y=1$, then eqn ① becomes

$$(-1+1)^n = nC_0 (-1)^0 1^n + nC_1 (-1)^1 1^{n-1} + nC_2 (-1)^2 1^{n-2} + \dots + nC_n (-1)^n 1^0$$

$$0 = nC_0 - nC_1 + nC_2 - \dots + (-1)^n nC_n$$

5. Let $A = \{a, b, c, d, e, f, g, h\}$, $B = \{1, 2, 3, 4, 5\}$. How many elements are there in $P(A \times B)$, the power set of $A \times B$?

A: No. of elements in A , $|A| = 8$
No. of elements in B , $|B| = 5$
No. of elements in $A \times B$, $|A \times B| = 8 \times 5 = 40$
No. of elements in $P(A \times B)$, $|P(A \times B)| = 2^{|A \times B|} = 2^{40}$

6. Does the formula $f(x) = \frac{1}{x^2 - 2}$ define

i) a function $f: \mathbb{R} \rightarrow \mathbb{R}$

ii) a function $f: \mathbb{Z} \rightarrow \mathbb{R}$?

A: i) $f: \mathbb{R} \rightarrow \mathbb{R}$; $f(x) = \frac{1}{x^2 - 2}$.

When $x = \sqrt{2} \in \mathbb{R}$; $f(x) = \frac{1}{(\sqrt{2})^2 - 2} = \frac{1}{0} \notin \mathbb{R}$

$\therefore f: \mathbb{R} \rightarrow \mathbb{R}$ is not a function under $f(x) = \frac{1}{x^2 - 2}$

ii) $f: \mathbb{Z} \rightarrow \mathbb{R}$; $f(x) = \frac{1}{x^2 - 2}$

Every element in $\mathbb{Z}$ has exactly one image.

$\therefore f: \mathbb{Z} \rightarrow \mathbb{R}$ is a function under $f(x) = \frac{1}{x^2 - 2}$.

7. Find the generating function for the sequence $1, 1, 1, \dots, 1, 0, 0, 0, \dots$ where the first $n+1$ terms are 1.

A: Generating function is $1 + x + x^2 + x^3 + \dots + x^n$.

$$= \frac{1 - x^{n+1}}{1 - x}$$

[Sum of n terms of a GP = $\frac{a(1 - r^n)}{1 - r}$. Here $a = 1$, $r = x$, $n = n+1$]

8. Find the coefficient of x^7 in the expansion of $(1+x+x^2+\dots)^{15}$.

A:
$$f(x) = (1+x+x^2+\dots)^{15}$$
$$= (1-x)^{-15}$$

$$(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \dots$$

$$(1-x)^{-15} = 1 + 15x + \frac{15 \cdot 16}{2!}x^2 + \frac{15 \cdot 16 \cdot 17}{3!}x^3 + \frac{15 \cdot 16 \cdot 17 \cdot 18}{4!}x^4 + \frac{15 \cdot 16 \cdot 17 \cdot 18 \cdot 19}{5!}x^5 + \frac{15 \cdot 16 \cdot 17 \cdot 18 \cdot 19 \cdot 20}{6!}x^6 + \frac{15 \cdot 16 \cdot 17 \cdot 18 \cdot 19 \cdot 20 \cdot 21}{7!}x^7 + \dots$$

Coefficient of $x^7 = \frac{15 \cdot 16 \cdot 17 \cdot 18 \cdot 19 \cdot 20 \cdot 21}{7!} = 116280$

9. Let $N = \{0, 1, 2, 3, \dots\}$. Define $f: N \rightarrow N$ as $f(m) = 3^m$. Show that f is a monoid homomorphism from $(N, +) \rightarrow (N, \cdot)$, where $(N, +)$, $(N, \cdot)$ are monoids under usual addition and multiplication respectively.

A:
$$f(m+n) = 3^{m+n} = 3^m \cdot 3^n = f(m) \cdot f(n)$$

$$f(0) = 3^0 = 1$$

$\therefore$ Monoid homomorphism.

10. Z_n is the group of integers mod n under modular addition. List the generators of (i) Z_{12} (ii) Z_p , where p is prime.

A: (i) $Z_{12} = \{0, 1, 2, \dots, 11\}$. Generators of Z_{12} are 1, 5, 7 and 11.

(ii) $Z_p = \{0, 1, 2, \dots, p-1\}$. Generators are 1, 2, 3, $\dots$, $p-1$.

PART B

Module I

11 (a) Prove the validity of the following argument:

If Rochelle gets the supervisor's position and works hard, then she will get a pay raise. If she gets the pay raise, then she will buy a new car. She has not purchased a new car. Therefore either Rochelle did not get the supervisor's position or she did not work hard.

A: Let p : Rochelle gets the supervisor's position

q : Rochelle works hard

r : Rochelle will get a pay raise.

s : Rochelle will buy a new car.

We have , $(p \wedge q) \rightarrow r$

$r \rightarrow s$

$\neg s$

$\therefore \neg p \vee \neg q$

Steps	Reasons
1) $r \rightarrow s$	premise
2) $\neg s$	premise
3) $(p \wedge q) \rightarrow r$	premise
4) $\neg r$	1, 2, Modus Tollens
5) $\neg (p \wedge q)$	3, 4, Modus Tollens
6) $\therefore \neg p \vee \neg q$	5, De Morgan's law

(b) Negate and simplify the statement: $\forall x [p(x) \rightarrow q(x)]$

A:

$$\begin{aligned}\neg \forall x [p(x) \rightarrow q(x)] &\Leftrightarrow \exists x \neg [p(x) \rightarrow q(x)] \\ &\Leftrightarrow \exists x \neg [\neg p(x) \vee q(x)] \\ &\Leftrightarrow \exists x [p(x) \wedge \neg q(x)]\end{aligned}$$

12(a) Establish the validity of the following argument by the method of proof by contradiction: $\neg p \leftrightarrow q, q \rightarrow r, \neg r$. Therefore p .

A:	Steps	Reasons.
	1) $\neg p \leftrightarrow q$	Premise
	2) $(\neg p \rightarrow q) \wedge (q \rightarrow \neg p)$	(1) and $\neg p \leftrightarrow q \Leftrightarrow (\neg p \rightarrow q) \wedge (q \rightarrow \neg p)$
	3) $\neg p \rightarrow q$	(2) conjunctive simplification
	4) $q \rightarrow \neg p$	(2) conjunctive simplification
	5) $q \rightarrow r$	Premise
	6) $\neg r$	Premise
	7) $\neg p$	Additional premise
	8) q	(7), (3) Modus Ponens
	9) r	(8), (5) Modus Ponens
	10) $r \wedge \neg r \Leftrightarrow F_0$	(9), (6) Rule of conjunction
	11) $\therefore p$	(1), (10) Proof by contradiction.

(b) Let $p(x)$, $q(x)$ and $r(x)$ be the following open statements.

$$p(x): x^2 - 7x + 10 = 0; \quad q(x): x^2 - 2x - 3 = 0; \quad r(x): x < 0$$

Determine the truth or falsity of the following statements, where the universe is all integers.

- | | |
|--|--|
| (i) $\forall x [p(x) \rightarrow \neg r(x)]$ | (ii) $\forall x [q(x) \rightarrow r(x)]$ |
| (iii) $\exists x [q(x) \rightarrow r(x)]$ | (iv) $\forall x [p(x) \rightarrow r(x)]$ |

A: $p(x): x^2 - 7x + 10 = 0$

$$x^2 - 5x - 2x + 10 = 0$$

$$x(x-5) - 2(x-5) = 0$$

$$(x-2)(x-5) = 0$$

$$x = 2, 5$$

$q(x): x^2 - 2x - 3 = 0$

$$x^2 + x - 3x - 3 = 0$$

$$x(x+1) - 3(x+1) = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3, -1$$

(i) True (If $x = 5$ or 2 then $x \geq 0$ is the statement)

(ii) False (If $x = 3$ or -1 , then $x < 0$ is the statement)
 $\therefore$ When $x = 3$, $x < 0$ is false

(iii) True (There exists an integer $x = 3$ or $x = -1$ such that $x < 0$)

(iv) False (False because when $x = 5, x = 2, x < 0$ is not possible)

Module 2

13.

- a) In how many ways can we distribute eight identical white balls into 4 distinct containers so that (i) no container is left empty. (ii) the 4th container has an odd no. of balls in it?
- b) An auditorium has a seating capacity of 800. How many seats must be occupied to guarantee that at least two people seated in the auditorium have the same first and last initials? (You may use pigeonhole principle).

ans:-

- a) (i) $n=4$.

It is given that no container is left empty.

$\therefore$ after distributing one ball to each container we've

$$r = 8 - 4 = 4$$

$$\therefore \text{required no. of ways} = {}^{n+r-1}C_r = {}^7C_4 = \frac{7!}{4!(7-4)!} = \underline{\underline{35}}$$

- (ii) It is given that the 4th container has an odd no. of balls.

$\therefore$ possible no. of balls in the 4th container are 1, 3, 5, 7.

$\therefore$ required no. of ways

$$= {}^{3+7-1}C_7 + {}^{3+5-1}C_5 + {}^{3+3-1}C_3 + {}^{3+1-1}C_1$$

$$= {}^9C_7 + {}^7C_5 + {}^5C_3 + {}^3C_1 = \underline{\underline{70}}$$

- b) No. of different initials = $26 \times 26 = \underline{\underline{676}}$

Hence in order to choose at least 2 people seated in the auditorium with same first and last initials = $26^2 + 1 = \underline{\underline{677}}$

$\therefore$ at least 677 seats must be occupied.

14. a) A woman has 11 friends among which 2 are married to each other. (i) In how many ways can she invite 5 of them to dinner? (ii) In how many ways can she invite 5 of them to dinner if the married couple attends the function together? (iii) In how many ways can she invite 5 of them to dinner if the married couple will not attend the function.
- b) Determine the coefficient of x^9y^3 in the expansion $(2x-3y)^{12}$.

ans:-

a) (i) $n=11$, $r=5$.

$$\therefore \text{no. of possible ways} = {}^{11}C_5 = \frac{11!}{5!(11-5)!} = \underline{\underline{462}}$$

(ii) $n=9$ ($\because$ the married couple attends the function)
 $r=3$

$$\therefore \text{no. of possible ways} = {}^9C_3 = \frac{9!}{3!(9-3)!} = \underline{\underline{84}}$$

(iii) $n=9$ ($\because$ the married couple will not attend the function)
 $r=5$

$$\therefore \text{no. of possible ways} = {}^9C_5 = \frac{9!}{5!(9-5)!} = \underline{\underline{126}}$$

b) The general term is, ${}^nC_r (2x)^{n-r} (-3y)^r$

$$\frac{12!}{9!3!} \cdot (2x)^9 \cdot (-3y)^3 = \frac{12!}{9!3!} 2^9 (-3)^3 x^9 y^3$$

$\therefore$ Coefficient of x^9y^3 is,

$$\underline{\underline{\frac{12!}{9!3!} \cdot 2^9 (-3)^3}}$$

Module-3

- 15 a) The relation R on $\mathbb{Z}^+$ is defined by aRb if 'a divides b'. Check whether R is i) reflexive ii) symmetric iii) transitive. Is R an equivalence relation?

Ans: i) R is reflexive if for all $a \in \mathbb{Z}^+$, aRa — (1)

Since $a|a$, aRa

$\therefore R$ is reflexive. — (1)

ii) R is symmetric if for all $a, b \in \mathbb{Z}^+$

$$aRb \Rightarrow bRa. \quad \text{--- (1)}$$

$$a|b \not\Rightarrow b|a$$

$$\text{eg: } 2|4 \text{ but } 4 \nmid 2$$

$\therefore R$ is not symmetric. — (1)

iii) R is transitive if for all $a, b, c \in \mathbb{Z}^+$,

$$aRb \text{ \& } bRc \Rightarrow aRc. \quad \text{--- (1)}$$

So let $a|b$ \& $b|c$

$$\text{ie, } b=ma \text{ \& } c=nb$$

$$c=nb$$

$$=n(ma)$$

$$\Rightarrow a|c$$

$$\Rightarrow aRc.$$

$\therefore R$ is transitive. — (1)

R is an equivalence relation if R is reflexive, symmetric \& transitive. — (1)

Since R is not symmetric, R is not an equivalence relation. — (1)

15b) Let $A = \{1, 2, 3\}$. Consider the relation R on A defined by $R = \{(1, 2), (2, 1), (2, 3)\}$. Is R symmetric? antisymmetric?

Ans: R is symmetric if for all $a, b \in A$,
 $aRb \Rightarrow bRa$. — ①

$$(2, 3) \in R \text{ but } (3, 2) \notin R \text{ — ①}$$

$\therefore R$ is not symmetric — ①

R is antisymmetric if for all $a, b \in A$,
 $aRb, bRa \Rightarrow a = b$. — ①

$$\text{In } R, (2, 1), (1, 2) \in R \text{ but } 1 \neq 2 \text{ — ①}$$

$\therefore R$ is not antisymmetric.

16a) Consider the set $A = \{a, b, c\}$. S.T. $\mathcal{Q}(A)$, the set of all proper subsets of A is a partially ordered set under the relation $\subseteq$, the set inclusion. Draw the Hasse diagram for the poset $(\mathcal{Q}(A), \subseteq)$. Is it a lattice?

Ans: $\mathcal{Q}(A) = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\} \}$ — ①

* Since each element is a subset of itself, it is reflexive — ①

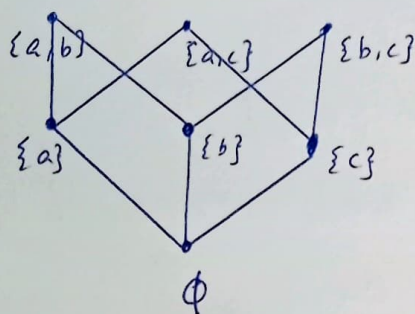
* For any $X, Y \in \mathcal{Q}(A)$, $X \subseteq Y \text{ \& } Y \subseteq X \Rightarrow X = Y$
 $\therefore \mathcal{Q}(A)$ is antisymmetric — ①

* For any $x, y, z \in Q(A)$,

Let $x \subseteq y$ & $y \subseteq z \Rightarrow x \subseteq z$.

$\therefore Q(A)$ is transitive — (1)

$\therefore Q(A)$ is a poset under the relation $\subseteq$.



— (3)

Since $\{a, b\}, \{a, c\}, \{b, c\}$ have no lub, it is not a lattice — (1)

16b) Define a distributive lattice. Give an example with justification.

Ans: A lattice $(A, +, \cdot)$ is said to be a distributive lattice if for any a, b, c in A ,

$$a + (b \cdot c) = (a + b) \cdot (a + c) \quad + \text{ is distributive over } \cdot$$

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \quad \cdot \text{ is distributive over } +$$

— (3)

eg: $(\mathbb{Z}^+, \leq)$ is a distributive lattice.

For the partial order relation, $\leq$

$$\text{lub}(a, b) = \max(a, b), \quad \text{glb}(a, b) = \min(a, b)$$

Let $a, b, c \in \mathbb{Z}^+$,

$$\begin{aligned}
 a \cdot (b+c) &= \min \{a, b+c\} \\
 &= \min \{a, \max(b, c)\} \\
 &= \max \{\min(a, b), \min(a, c)\} \\
 &= \max \{a \cdot b, a \cdot c\} \\
 &= (a \cdot b) + (a \cdot c)
 \end{aligned}$$

$$\begin{aligned}
 a + (b \cdot c) &= \max \{a, b \cdot c\} \\
 &= \max \{a, \min\{b, c\}\} \\
 &= \min \{\max\{a, b\}, \max\{a, c\}\} \\
 &= \min \{a+b, a+c\} \\
 &= (a+b) \cdot (a+c)
 \end{aligned}$$

← (3)

Module - 4

- 17.
- Solve the recurrence relation $a_{n+2} + a_n = 0$; $n \geq 0, a_0 = 0, a_1 = 3$.
 - Determine the sequence generated by the exponential generating function $f(x) = \frac{1}{1-x}$.

ans:-

a) characteristic equation is, $x^2 + 1 = 0$

$$\rightarrow x^2 = -1 \Rightarrow x = \pm i.$$

$\therefore$ general solution is, $a_n = A_1(i)^n + A_2(-i)^n$.

Consider $z = i$, $y = 1$, $z = 1 (\cos \pi/2 + i \sin \pi/2) = \underline{\underline{i \sin \pi/2}}$

Consider $z = -i$, $y = -1$

$$z = |-1| \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) = i \sin \frac{3\pi}{2} = -i \sin \frac{\pi}{2}$$

$$a_n = A_1 \left(i \sin \frac{\pi}{2} \right)^n + A_2 \left(-i \sin \frac{\pi}{2} \right)^n$$

$$= A_1 i \sin \frac{n\pi}{2} + A_2 \left(-i \sin \frac{n\pi}{2} \right)$$

$$= i(A_1 - A_2) \sin \frac{n\pi}{2} = B \sin \frac{n\pi}{2} ; \text{ where } B = (A_1 - A_2)i$$

$$a_0 = 0$$

$$\Rightarrow 0 = B \sin 0$$

$$\left\{ \begin{array}{l} a_1 = 3 \\ \Rightarrow 3 = B \sin \frac{\pi}{2} \\ \Rightarrow \underline{\underline{B = 3}} \end{array} \right.$$

$\therefore$ general solution is, $\underline{\underline{a_n = 3 \sin \frac{n\pi}{2} ; n \geq 0}}$

$$b) f(x) = \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$= 1 + x + \frac{2! x^2}{2!} + \frac{3! x^3}{3!} + \dots$$

$\therefore$ sequence is, $0!, 1!, 2!, 3!, \dots$

18.

a) Solve the recurrence relation $a_{n+2} - 10a_{n+1} + 21a_n = 7 \cdot 11^n$

b) Find the unique solution of the recurrence relation,

$$2a_n - 3a_{n-1} = 0 ; n \geq 1, a_4 = 81$$

ans: To find $a_n^{(h)}$

a) Characteristic equation is, $x^2 - 10x + 21 = 0$

$$\rightarrow x = 3, 7$$

$$\therefore a_n^{(h)} = \underline{\underline{A_1 3^n + A_2 7^n}}$$

To find $a_n^{(p)}$

$$RHS = 7 \cdot 11^n$$

Here 11 is not a root of the characteristic equation.

$$a_n^{(p)} = B \cdot 11^n$$

Substituting it in the given recurrence relation, $a_{n+2} - 10a_{n+1} + 21a_n = 7 \cdot 11^n$

$$B \cdot 11^{n+2} - 10B \cdot 11^{n+1} + 21 \cdot B \cdot 11^n = 7 \cdot 11^n$$

$$\text{Finding } B = \frac{7}{32}$$

$$\therefore a_n^{(p)} = \frac{7}{32} \cdot 11^n$$

$$\therefore \text{General solution is, } a_n = A_1 3^n + A_2 7^n + \underline{\underline{\frac{7}{32} \cdot 11^n}}$$

b) characteristic equation is, $2x - 3 = 0 \Rightarrow x = \frac{3}{2}$

$$\therefore \text{general soln is, } a_n = A \cdot \left(\frac{3}{2}\right)^n$$

$$\text{Given } a_4 = 81$$

$$\Rightarrow a_4 = A \cdot \frac{3^4}{2} = 81$$

$$\Rightarrow A = 16$$

$$\therefore \text{general solution is, } a_n = \underline{\underline{16 \times \left(\frac{3}{2}\right)^n}}; n \geq 0$$

Module 5

19.

- a) Define (i) Semigroup (ii) monoid (iii) group. Give one example each, different from one another. Is $\mathbb{R}$, the set of real numbers, a group under multiplication? Justify.
- b) If H and K are sub-groups of a group G , prove that HK is also a sub-group of G .

ans:-

a) (i) Semi-group.

If S is a non-empty set and $*$ is a closed binary operation on S , then the algebraic system $(S, *)$ is called a semi-group if the operation $*$ is associative.

eg:- $(\mathbb{Z}^+, +)$

(ii) Monoid

It is a semi-group with an identity element e . It is denoted by $(M, *, e)$

eg:- Let $N = \{0, 1, 2, \dots\}$. Then $(N, +)$ is a monoid with identity element 0 .

(iii) Group

If G is a non-empty set and $*$ is a binary operation on G , then $(G, *)$ is called a group if the following conditions are satisfied:

* Closure property :- $\forall a, b \in G, a * b \in G$

* Associative property :- $\forall a, b, c \in G, a * (b * c) = (a * b) * c$

* Existence of an identity :- $\exists e \in G$ with $a * e = a = e * a$

* Existence of inverse :- $\forall a \in G, \exists a^{-1} \in G$ s.t. $a * a^{-1} = e = a^{-1} * a$.

eg:- $(\mathbb{R}, +), (\mathbb{Q}, +)$.

$(\mathbb{R}, \cdot)$ is not a group.

Since 0 has no inverse.

b) Since H and K are subgroups of G , $e \in H$ and $e \in K$
 $\Rightarrow e \in H \cap K$.

$\therefore$ identity element exist.

Let $x, y \in H \cap K \Rightarrow x, y \in H$ and $x, y \in K$
 $\Rightarrow xy \in H$ and $xy \in K$
 $\Rightarrow xy \in H \cap K$.

$\therefore$ closure property is satisfied.

Let $a \in H \cap K \Rightarrow a \in H$ and $a \in K$
 $\Rightarrow a^{-1} \in H$ and $a^{-1} \in K$
 $\Rightarrow a^{-1} \in H \cap K$.

$\therefore$ inverse exist.

Also $\forall x, y, z \in H \cap K$, $x + (y + z) = (x + y) + z$.

Since $x, y, z \in H$ and $x, y, z \in K$ and H and K are subgroup

$\therefore$ associative property is satisfied.

$\therefore H \cap K$ is a sub-group of G .

20.

a) Let $(G, \circ)$ and $(H, *)$ be groups with respective identities e_G, e_H . If $f: G \rightarrow H$ is a homomorphism, prove that for all $a \in G$ and $n \in \mathbb{Z}$

(i) $f(e_G) = e_H$ (ii) $f(a^{-1}) = [f(a)]^{-1}$ (iii) $f(a^n) = [f(a)]^n$.

b) Let $G = (\mathbb{Z}, +)$ be the group of integers under addition.

Let $H = \{\dots, -8, -4, 0, 4, 8, \dots\}$. Show that H is a subgroup of G . write all left cosets of H in G .

ans:-

$$\begin{aligned} a) (i) e_H * f(e_G) &= f(e_G) \quad (\because e_H \text{ is the identity element in } H) \\ &= f(e_G \cdot e_G) \quad (\because e_G \text{ is the identity element in } G) \\ &= f(e_G) * f(e_G) \\ \Rightarrow \underline{e_H} &= \underline{f(e_G)} \quad (\text{by cancellation law}) \end{aligned}$$

(ii) Let $a, a^{-1} \in G$

$$a \cdot a^{-1} = e_G$$

$$f(a \cdot a^{-1}) = f(e_G)$$

$$f(a) * f(a^{-1}) = e_H$$

$\Rightarrow f(a^{-1})$ is the inverse of $f(a)$.

$$\therefore \underline{f(a^{-1}) = [f(a)]^{-1}}$$

(iii) $\forall a \in G$, by closure property,

$$a \cdot a \in G, a \cdot a \cdot a \in G, \dots, a^n = a \cdot a \dots a \in G$$

$$f(a \cdot a \dots n \text{ times}) = f(a) * f(a) * \dots n \text{ times}$$

$$\therefore \underline{f(a^n) = [f(a)]^n}$$

b) H is closed under addition, H has the identity element 0 and every element in H has inverse.

$\therefore H$ is a sub-group of G .

Left cosets are,

$$0 + H = H$$

$$1 + H = \{ \dots, -7, -3, 1, 5, 9, \dots \}$$

$$2 + H = \{ \dots, -6, -2, 2, 6, 10, \dots \}$$

$$3 + H = \{ \dots, -5, -1, 3, 7, 11, \dots \}$$